TERMODINÁMICA APLICADA · FORMULARIO

Formulario de una página

Constantes

$$R_{\text{aire}} = 0{,}287\ \tfrac{\text{kJ}}{\text{kg·K}} \quad c_p = 1{,}005\ \tfrac{\text{kJ}}{\text{kg·K}} \quad c_v = 0{,}718\ \tfrac{\text{kJ}}{\text{kg·K}} \quad k = 1{,}4 \quad R_u = 8{,}314\ \tfrac{\text{kJ}}{\text{kmol·K}}$$

Agua · gases · sistemas

$$x = \frac{v - v_f}{v_g - v_f} \qquad PV = mRT \qquad Q - W = \Delta U \qquad W = \frac{P_2V_2 - P_1V_1}{1-n}$$

$$\dot{Q} - \dot{W} = \dot{m}\left[\Delta h + \frac{\Delta c^2}{2} + g\Delta z\right] \qquad \eta_T = \frac{h_1 - h_2}{h_1 - h_{2s}}$$

Segundo principio · entropía

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H} \qquad \text{COP}_{\text{ref}} = \frac{Q_C}{W} \qquad \Delta s = c_p \ln\frac{T_2}{T_1} - R\ln\frac{P_2}{P_1} \qquad S_{\text{gen}} \geq 0$$

Ciclos

$$\eta_{\text{Brayton}} = 1 - r_p^{-(k-1)/k} \qquad \eta_{\text{Otto}} = 1 - r^{1-k} \qquad \text{COP} = \frac{h_1 - h_4}{h_2 - h_1}$$

Psicrometría · mezclas

$$\omega = 0{,}622\frac{p_v}{P - p_v} \qquad \text{HR} = \frac{p_v}{p_g} \qquad h = 1{,}006T + \omega(2501 + 1{,}86T) \qquad \Delta S = -R_u \sum n_i \ln y_i$$

Combustión

$$\text{A/F}_{\text{CH}_4} = 17{,}19\ \text{kg/kg} \qquad \text{PCS} - \text{PCI} = \text{agua condensada} \cdot h_{fg} \qquad y_{\text{H}_2\text{O}} = \frac{n_{\text{H}_2\text{O}}}{n_{\text{productos}}}$$